Nat Hazards (2015) 75:2847–2862
DOI 10.1007/s11069-014-1464-1

ORIGINAL PAPER

Numerical analysis of tsunami–structure interaction using a modified MPS method

Yu Huang · Chongqiang Zhu

Received: 26 July 2014 / Accepted: 3 October 2014 / Published online: 12 October 2014

Abstract Coastal structures are regarded as an effective defense for protecting the coastal area from tsunami disasters, considering that the tsunamis cannot be predicted. Therefore, much attention should be focused on tsunami–structure interaction (TSI). We must determine dynamic characteristics of the TSI such as wave height, free-surface elevation, dynamic wave pressure, and overtopping volume, all of which are essential to the design of coastal structures. The traditional mesh-based numerical method fails to accurately model TSI, because of large deformations that cause numerical diffusion. The moving particle simulation (MPS) is a pure Lagrangian mesh-less method, which can track free-surfaces with large deformations. The original MPS suffers from serious pressure fluctuations that affect the accuracy of the simulation. We modified three aspects of the original MPS: the kernel function, the source term, and the search of free-surface particles. We verified that these changes improved the pressure stability using two benchmark problems. Then, we applied the modified MPS to simulate the TSI, using a solitary wave to model the tsunami. We quantitatively and qualitatively compared our numerical results with the experimental data. The numerical results were consistent with the experimental data, which indicates that the modified MPS can capture the essential dynamic characteristics of the TSI and reproduce the entire interaction between the tsunami and structure.

Keywords Tsunami · MPS · Coastal structure · Pressure fluctuation · Interaction

Y. Huang (✉) · C. Zhu
Department of Geotechnical Engineering, College of Civil Engineering, Tongji University, Shanghai 200092, China
e-mail: yhuang@tongji.edu.cn

Y. Huang
Key Laboratory of Geotechnical and Underground Engineering of the Ministry of Education, Tongji University, Shanghai 200092, China

1 Introduction

Tsunamis are a series of destructive waves caused by large-scale sudden movements of the seabed due to earthquakes, submarine slides, volcanic eruptions, or explosions (Lau et al. 2011). A tsunami typically lasts for 100–2,000 s and has a velocity of 600–900 km/h, with wavelengths up to 5×10^5 m in the open ocean (Ward 1989). The massive energy remains almost unchanged as the tsunami approaches shores, which is catastrophic to humans and causes serious damages to coastal structures. For example, the death toll reached 20,416 (including missing people) in the March 11, 2011, Great East Japan Tsunami, which was trigged by a magnitude 9.0 (M_w) undersea megathrust earthquake (Arikawa et al. 2012). The coastal area of China, whose coastline exceeds 18,000 km (State Council Information Office of China 1998), is an economically advanced area with a large population. The land area of the coastal regions represents only 17 % of the total area of China, but it supports approximately 42 % of the national population and accounts for more than 73 % of the GDP (National Natural Science Foundation of China 2011). The South China Sea is over 1,200 m deep in some places, and its tectonic configuration is generated by interactions between the Indian Ocean plate, the Eurasia plate, and the Philippine Sea plate (Lin et al. 1980). It is regarded as the zone with the highest tsunami risk in China (Zhou and Adams 1988). Therefore, the coastal area must be protected from tsunamis to achieve sustainable development. However, it is impossible to predict a tsunami before an under-ocean earthquake actually happens (Titov et al. 2005). Fortunately, the coastal area is protected by a seawall, which is regarded as an effective countermeasure for tsunami disasters (Fujima 2006). Thus, there is an urgent need for research into TSI, to determine the dynamic characteristics of coastal structures. These characteristics could provide preliminary advice for designers to optimize their scheme and, in turn, could protect coastal areas from disasters.

The methods used to study the interaction between coastal structures and tsunamis fall into three categories: field disaster investigation, hydraulic model experiments, and numerical simulations. These methods have their own advantages and disadvantages. Field disaster investigation directly evaluates the degree of seawall damage and collects raw data of tsunamis. However, it cannot reproduce the interaction processes between tsunamis and seawalls, and the data are rare and inaccurate because the evidence can quickly disappear (Farreras 2000). Hydraulic model experiments can precisely reproduce the actual physical phenomena of TSI. However, it is expensive and time-consuming to set seabed configurations, manufacture seawall models, and record a variety of data such as wave height, wave pressure, and overtopping volume (Hanzawa et al. 2012). Fortunately, developing computer technologies have led to various sophisticated numerical methods that simulate complicated TSI problems. Numerical simulations are powerful and efficient and have many advantages over the other methods. Hence, numerical simulations have a significant role in analyzing the TSI.

A tsunami wave can be characterized as large scale and transient, with complex free-surface behaviors on geometrically complex topography (Cleary and Prakash 2004). TSI research typically involves measuring the wave height, pressure on the seawall, and overtopping volume. Thus, the numerical model must accurately track the free-surface (including the tsunami wave propagation, fragmentation, and splashing) and precisely predict the wave force and overtopping volume, which are essential for designing the coastal structures. Previous studies (Garcia et al. 2004; Ohyama and Nadaoka 1992; Lin 2004; Losada et al. 2008) used a mesh method to simulate the TSI. They achieved some promising results. However, the traditional mesh method suffers from numerical diffusion when it is applied to a TSI with a large deformation, resulting in excessive mesh twisting and distortion. Fortunately, the rapidly developing mesh-free method has significant advantages over traditional mesh-based methods for these types of TSI. These include accurate tracking of the free-surface and precise predictions of the overtopping volume.

MPS represents liquids (or solids) using a number of arbitrarily distributed particles that contain field variables such as mass, density, pressure, and velocity. The terms in the governing equation are described by interactions of the neighboring particles. MPS is particularly well suited to TSI simulations. MPS was originally developed by Koshizuka and Oka (1996) for the fragmentation of incompressible fluids and has been successfully applied to various engineering fields. The growth of bubble in the transient pool boiling is studied using MPS method (Heo et al. 2002). MPS method is applied to investigate the microscopic behavior of blood flow (Tsubota et al. 2006). The adhesion of cells in micro-channels is simulated using MPS method (Suzuki et al. 2006). Huang and Zhu (2014) innovatively introduced the Bingham constitutive model combined with the Mohr–Coulomb failure criterion to analyze the dynamic characteristics of flow slide in municipal solid waste dumps.

Some previous studies have used the particle method for TSI: a simulation of a wave breaking and overtopping process at an upright seawall (Hayashi et al. 2001), overtopping volume at a vertical seawall (Gotoh et al. 2005), a numerical analysis of the wave interaction with partially immersed breakwater using the smoothed particle hydrodynamics (SPH) combined with a large eddy simulation (LES) method (Gotoh et al. 2004), and a calculation of the movement of caisson breakwater using the SPH method (Rogers et al. 2010). These previous studies did not systematically research TSI and were not quantitative. Moreover, dynamic pressure is essential to the design of coastal structures, but is rarely determined using the particle method because of excessive numerical fluctuations in the pressure. In particle methods, the position of particle is updated by particles pushing away each other. In other words, the pressure fluctuation is caused by the excessive concentration of particle produces repulsive forces (Gotoh et al. 2005). Unfortunately, the pressure oscillation has not been fully solved until now. However, some schemes have been proposed to improve the stability of pressure and have got many promising results (Khayyer and Gotoh 2009; Tanaka and Masunaga 2010; Kondo and Koshizuka 2011). Therefore, in this paper, the original MPS code was revised in three aspects (the kernel function, the source term, and the search of free-surface particles) to improve the pressure stability. We verified the modified MPS by simulating two representative models: the dam break problem and the static pressure problem. Then, we quantitatively and qualitatively investigated the wave height, dynamic pressure on the seawall, free-surface shape, and overtopping volume using the modified MPS, to reproduce the entire TSI process. Our simulation results agreed well with experimental results from published literatures.

2 Outline of the MPS method

2.1 Governing equations

The continuity equation is defined as

$$\frac{\mathrm{D}\rho}{\mathrm{D}t} = 0, \tag{1}$$

and the Navier–Stokes is

$$\frac{\mathrm{D}\mathbf{u}}{\mathrm{D}t} = -\frac{1}{\rho}\nabla P + \nu\nabla^2\mathbf{u} + \mathbf{g}, \tag{2}$$

where ρ is the density, t is the time, $\mathbf{u}$ is the velocity vector, ∇ is the gradient, P is the pressure, ν is the kinematic viscosity, ∇^2 is the Laplacian, and $\mathbf{g}$ is the gravitational acceleration vector.

2.2 MPS discretization

The gradient and Laplacian at particle i are discretized using a local weighted average particle interaction model, which is defined as (Koshizuka and Oka 1996):

$$\langle\nabla\phi\rangle_i = \frac{d}{n^0}\sum_{i\neq j}\left[\frac{\phi_j - \widehat{\phi}_i}{\left|\boldsymbol{r}_i - \boldsymbol{r}_j\right|^2}\left(\boldsymbol{r}_i - \boldsymbol{r}_j\right)w(|\boldsymbol{r}_j - \boldsymbol{r}_i|)\right], \tag{3}$$

$$\left\langle\nabla^2\phi\right\rangle_i = \frac{2d}{\lambda n^0}\sum_{j\neq i}\left[(\phi_j - \phi_i)w(|\boldsymbol{r}_j - \boldsymbol{r}_i|)\right], \tag{4}$$

and

$$\lambda = \frac{\sum_{j\neq i} w(|\boldsymbol{r}_j - \boldsymbol{r}_i|)|\boldsymbol{r}_j - \boldsymbol{r}_i|^2}{\sum_{j\neq i} w(|\boldsymbol{r}_j - \boldsymbol{r}_i|)}, \tag{5}$$

where d is the number of dimensions, $\widehat{\phi}_i$ is the minimum value in the interaction area, and $\boldsymbol{r}$ is the particle position vector. n^0 is the initial value of the particle number density (PND), which is defined as

$$n_i = \sum_{j\neq i} w\left(\left|\boldsymbol{r}_j - \boldsymbol{r}_i\right|\right), \tag{6}$$

where w is the kernel function of the distance between two particles $r = \left|\boldsymbol{r}_j - \boldsymbol{r}_i\right|$. In this paper, we use a new kernel function (Jung et al. 2008) instead of the one used in the original MPS. This is because of the singularity of the original kernel function at $r = 0$, which leads to pressure oscillations. The new kernel function is

$$w(r) = \begin{cases} \left(1 - \dfrac{r}{r_\mathrm{e}}\right)^3\left(1 + \dfrac{r}{r_\mathrm{e}}\right)^3 & (0 \le r < r_\mathrm{e}) \\ 0 & (r \ge r_\mathrm{e}) \end{cases}, \tag{7}$$

where r_e is the radius of the particle interaction area, which is 2.1 times the initial particle spacing.

2.3 Calculation process

The time integration process in each time step is divided into two parts: prediction and revision. In the first part, the predicted velocity ($\mathbf{u}_i^*$) is explicitly calculated by considering the given external force and gravity terms. Then, the predicted position of particle i ($\mathbf{r}_i^*$) is calculated using

$$\mathbf{r}_i^* = \mathbf{r}_i^k + \Delta t\mathbf{u}_i^*, \tag{8}$$

where $\mathbf{r}_\mathbf{i}^k$ is the position of particle i in step k, and Δt is the time increment. When the particle position is updated, the predicted PND (n_i^*) is also automatically updated, which means that the mass is not conserved (i.e., $n_i^* \neq n^0$).

In the second part, the pressure (which is used to revise the velocities and positions of the particles) is implicitly calculated by solving the Poisson equation of pressure. The Poisson equation in the original MPS is (Koshizuka and Oka 1996)

$$\left\langle \nabla^2 P^{k+1} \right\rangle_i = \frac{\rho}{\Delta t^2} \frac{n_i^* - n^0}{n^0}. \tag{9}$$

The right-hand side of Eq. (9) keeps the PND constant, so that it can satisfy the incompressibility in the simulation. Although the source term used in the original MPS avoids accumulations of density errors, it causes high-frequency pressure fluctuations in both time and space. Previous studies have proposed some valuable methods for improving the pressure stability from the point of view of the source term. Khayyer and Gotoh (2009) used a source term that is a function of the velocity divergence, which reduced the pressure oscillations to some degree. However, this form of the source term leads to density error accumulations during the computation. A hybridized source term was proposed by Tanaka and Masunaga (2010), which performed better than the original MPS. However, the hybridized source term does not have a definite physical meaning, because it is not derived from the hydrodynamic equation. In this study, we used the source term proposed by Kondo and Koshizuka (2011) with one main part and two error-compensating parts. It is defined as

$$\frac{1}{\rho} \langle \nabla^2 P^{k+1} \rangle_i = \frac{n_i^* - 2n_i^k + n_i^{k-1}}{n^0 \Delta t^2} + \frac{a}{\Delta t} \frac{n_i^k - n_i^{k-1}}{n^0 \Delta t} + \frac{b}{\Delta t^2} \frac{n_i^k - n^0}{n^0}, \tag{10}$$

where a and b are dimensionless coefficients for the error-compensating parts and are set to 0.5 and 0.05, respectively, by performing sufficient numerical experiments.

2.4 Free-surface judgment

The free-surface is recognized based on the condition of the PND as follows:

$$n_i^* < \beta n^0, \tag{11}$$

where β is a threshold parameter, set to $\beta = 0.9$ in this study. In the original MPS, $\beta = 0.97$, which causes pressure instabilities because some inside particles are recognized as free-surface particles. The pressure of the misjudged particles is set to zero, which results in pressure fluctuations in space.

3 Solitary wave generation

A solitary wave (which belongs to the gravity wave) is used to study tsunamis, because it has a similar propagation behavior to the tsunami wave (Synolakis and Bernard 2006). Boussinesq (1872) first proposed the solitary wave profile equation and solitary wave velocity equation in constant depths, by solving the linear shallow water equations. Goring (1979) presented the nonlinear trajectory theory of a paddle-generated solitary wave. We generated a solitary wave in this research, which can be depicted by the following equations.

The solitary wave profile equation is

$$\eta(x,t) = H\ \mathrm{sech}^2[k(ct - x)], \tag{12}$$

where

$$c = \sqrt{g(H + h)} \tag{13}$$

and

$$k = \sqrt{\frac{3H}{4h^3}}. \tag{14}$$

Here, H is the wave height, c is wave velocity, x is the position of the wave maker, and h is the water depth.

The wave maker stroke S is

$$S = \frac{2H}{kh} = \sqrt{\frac{16H}{3h}}h, \tag{15}$$

and the duration (T) of the wave maker motion is

$$T = \frac{2 \times \left(3.80 + \frac{H}{h}\right)}{kc}. \tag{16}$$

Equations (15) and (16) generate the solitary wave expressed in Eq. (12).

The position of the paddle X at time t is described as

$$X(t) = \frac{H}{kh}\tanh[k(ct - X(t))]. \tag{17}$$

The original point of the wave maker paddle is set to ($-T/2$, $-S/2$), to make the paddle move forward at the initial time 0. The position of the paddle is

$$X(t)=\frac{H}{kh}\left\{1 + \tanh k\left[c\left(t - \frac{T}{2}\right) - \left(X(t) - \frac{S}{2}\right)\right]\right\}. \tag{18}$$

During the wave making process of the numerical simulation, the paddle moves forward according to Eq. (18) and then remains static.

As an example, Fig. 1 shows the paddle displacement and velocity distribution used to produce a 0.09 m high solitary wave in water that is 0.2 m deep.

4 Verification and validation of the modified MPS method

In a previous paper (Huang and Zhu 2014), we verified and validated the modified MPS method using two sample problems (the dam break and static pressure problems). The simulation results were compared with experimental data or analytical results, which indicated that the modified MPS can effectively reduce the pressure fluctuation. Moreover, it can improve some macro-behaviors, such as the behavior of the water front in the dam break problem. Therefore, the accuracy and improvement of the modified MPS have been demonstrated for dynamic pressure predictions.

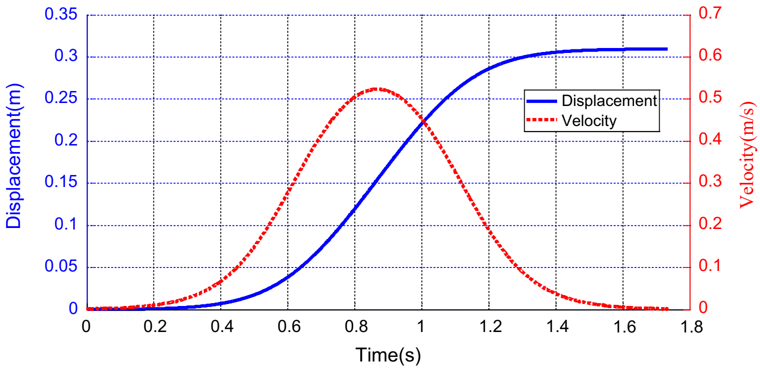


Fig. 1 Time series of the velocity and displacement of the wave maker paddle

5 Application

5.1 Experimental conditions

Hsiao and Lin (2010) conducted a set of laboratory experiments to model three types of tsunamis interacting with an impermeable seawall on a sloping beach. They considered a bore, an impinging wave, and an overtopping wave. The experiments were carried out at the Tainan Hydraulics Laboratory. The wave flume was 22 m long with a depth of 0.75 m and a width of 0.5 m. A solitary wave was generated using a piston-type wave maker, by applying the Goring mathematical model mentioned above. In the experiment, the tsunami wave approached a beach with a slope of 1:20 and ran up against an impermeable seawall. This wall had a seaward slope of 1:4 and a landward slope of 1:1.8, starting at a distance of 3.6 m from the toe of beach. The free-surface deformation was recorded by wave gauges along the flume, and the pressure was measured by pressure transducers along the seawall. A sketch of the wave flume layout and pressure transducer locations is shown in Fig. 2.

5.2 Computational conditions

There were three types of solitary waves in the experiment. Type 1 was a turbulent bore, Type 2 was a wave that collapsed on the seawall and then overtopped it, and Type 3 was a wave that directly overtopped the crest. In this paper, we only studied the first case. The water depth (h_0) was 0.2 m, and the wave height (H) was 0.07 m. We reduced the flat bed in the computational domain to 2.5 m because of computational limitations, which was longer than the wavelength of the solitary wave. Koshizuka et al. (2004) reported that one wavelength was sufficient for the flume length in shipping water analysis. Gotoh et al. (2005) also analyzed the overtopping discharge using a shorter model. The offshore wave that was generated by the wave maker set at the end of the horizontal bed was controlled so that the wave height at reference gauge $g2$ (placed at the beach toe) coincided with that of the experimental results. In the MPS method, the viscous term is used to describe the energy dissipation of laminar flow. Shao and Lo (2003) adopted a constant viscosity to simulate breaking wave without employing an appropriate turbulence model and got credible results especially after the stable breaking wave had formed. Hence, in the simulation, we used a constant viscosity to deal with the energy dissipation caused by breaking wave.

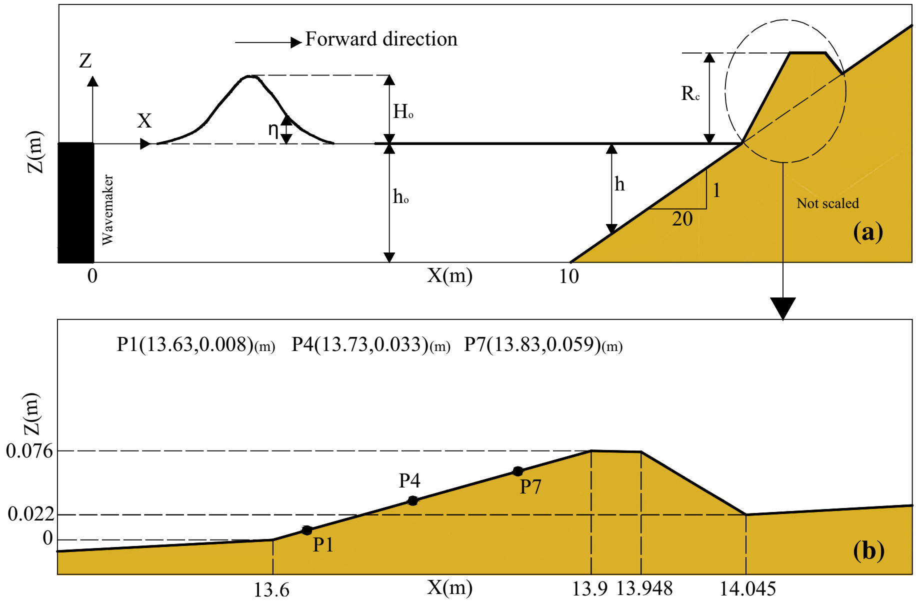


Fig. 2 **a** Schematic diagram of the flume layout. **b** Schematic diagram of the location of each pressure transducer (based on Hsiao and Lin 2010)

The wave flume, wave maker, and seawall were represented by fixed particles, and the water was represented by fluid particles. All the particles were placed in the regular form, with an initial distance between adjacent particles (d_0) of 0.02 m. The water density was 1,000 kg/m^3, and the gravity acceleration was 9.8 m/s^2. The total number of particles was 2,865, which included 1,742 fluid particles, 359 P-wall particles, and 764 wall particles. The calculation took about 80 min on a personal computer with 2.7 GHz CPU and 2 GB RAM.

6 Results and discussion

In this section, we qualitatively and quantitatively compare the numerical results and experiment data, including the wave height, free-surface, dynamic pressure load, and overtopping discharge.

6.1 Wave height

The wave height was measured in the experiment by setting wave gauges along the flume. In the numerical simulation, we recorded the wave height at the corresponding locations. Figure 3 shows the time series of the wave height at $g2$, $g10$, $g22$, and $g28$. The red line represents the experimental result, and the blue line represents the calculated result. The wave height at $g2$ agrees well with the experimental results, which indicates that the solitary wave generated by the wave maker was accurate and effective. The reflected wave, whose maximum wave height agrees with the physical model, which was approximately 3 mm at about 4.8 s, is observed in Fig. 3b. The wave height occasionally oscillates in the

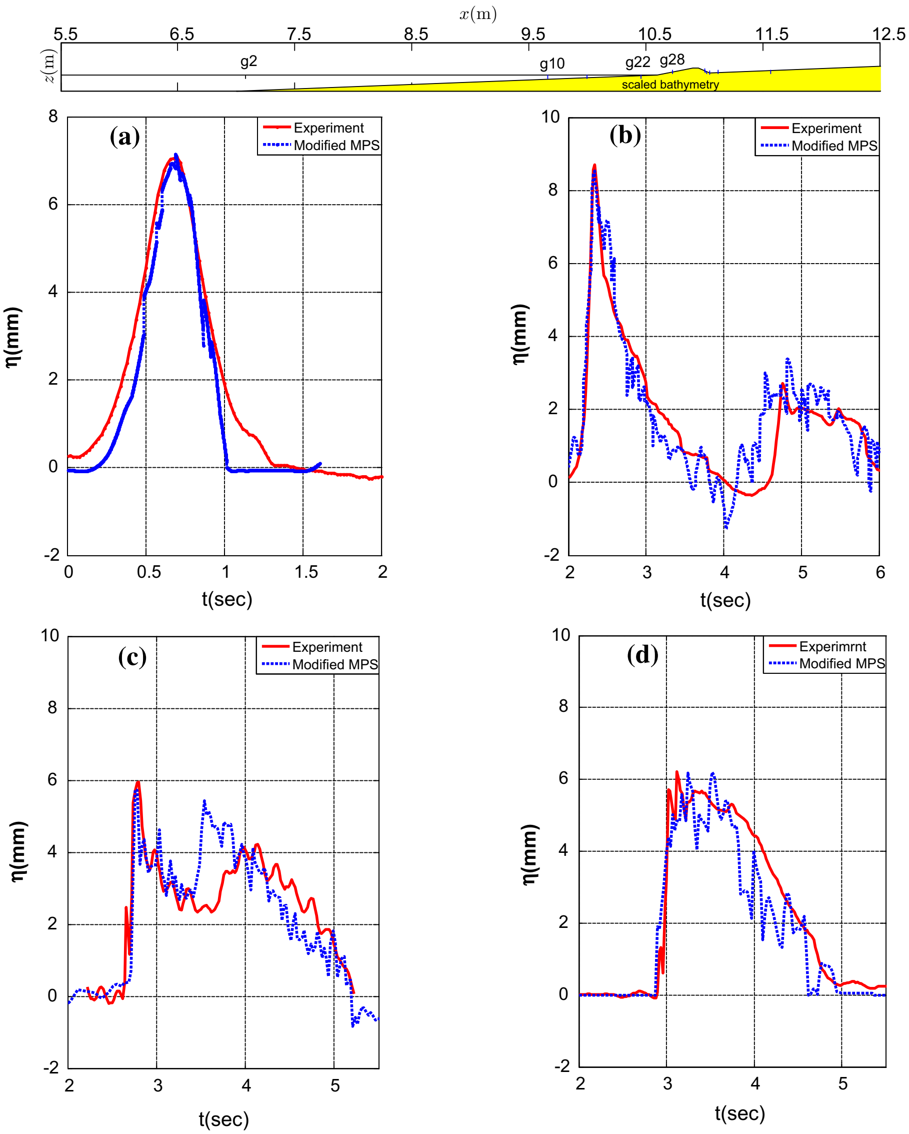


Fig. 3 Time series of the wave height at **a** $g2$; **b** $g10$; **c** $g22$; and **d** $g28$ (the experimental data were provided by Hsiao and Lin 2010)

numerical simulation because of the particle's random motion. In general, the modified MPS captured the wave propagation process.

6.2 Free-surface elevation

The solitary wave propagated along the channel and then impacted and overtopped the seawall. Figure 4 quantitatively and qualitatively shows the impact and overtopping process. The experimental pictures, measured data, and numerical results are plotted together for comparison. The numerical results agree well with the experiment. At $t = 2.89$ s, the tsunami wave arrived at the toe of the seawall. At $t = 3.01$ s, the turbulence bore began to impact the seawall. The wave height in the calculation is very close to the height in the experiment. The overtopping process is well reproduced in Fig. 4c–e, and the simulation results are accurate. However, there are some discrepancies between the experiments and numerical simulations. There is time backwardness, the experiment profile is not fully occupied by fluid particles, and some particles are out the profile. The phenomena may be partly due to the unpredictability of breaking waves, and the ignorance of the surface tension in the modified MPS simulation. The uncertain entrapped bubbles in the iterative experiment also cause disagreements. However, the comparison of numerical results and experiment data demonstrates that the modified MPS method is an effective tool for reproducing the overtopping process.

6.3 Dynamic wave pressure

Figure 5 compares the computed dynamic wave pressures (blue line) with corresponding experimental data (red line). The computed dynamic wave pressure got from the original MPS, which suffers from serious oscillation, is also presented. The maximum of the dynamic wave pressure is nearly ten times as much as the experimental data, which is consistent with previous works (Huang and Zhu 2014; Lee et al. 2011). The pressure distribution of the numerical simulation agrees with the research reported by Bullock et al. (2001). The initial impact pressure is a sharp peak and decrease, with a secondary boost and then a drop to zero. This comparison demonstrates that the modified MPS method can model typical features of the pressure time series when the tsunami impacts the seawall, with acceptable deficiencies.

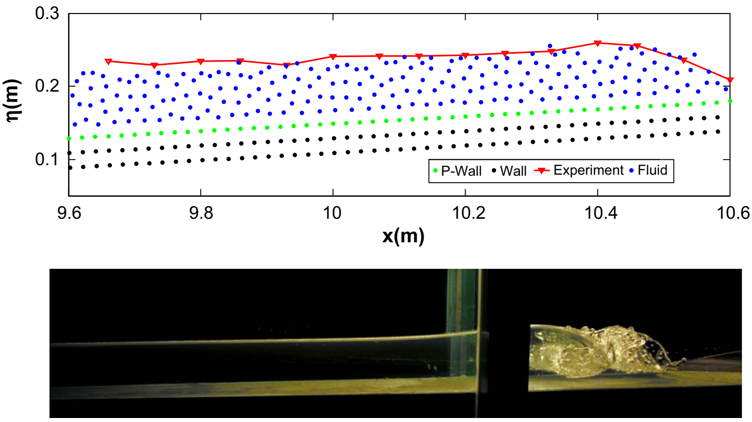
The dynamic wave force was calculated by integrating the pressure spread across the seaward part of the seawall and is shown in Fig. 6. The red line represents the data recorded during the physical model tests, and the blue line represents the numerical results. We found that the maximum wave force was approximately 150 N/m, both in the experiment and simulation, and occurred at almost the same time (about 3.2 s). This indicates that the modified MPS method can reproduce the force on the seawall during the tsunami wave interaction. It can be concluded that the modified MPS method provides reliable information for designing coastal structures and computing the dynamic wave force.

6.4 Overtopping volume

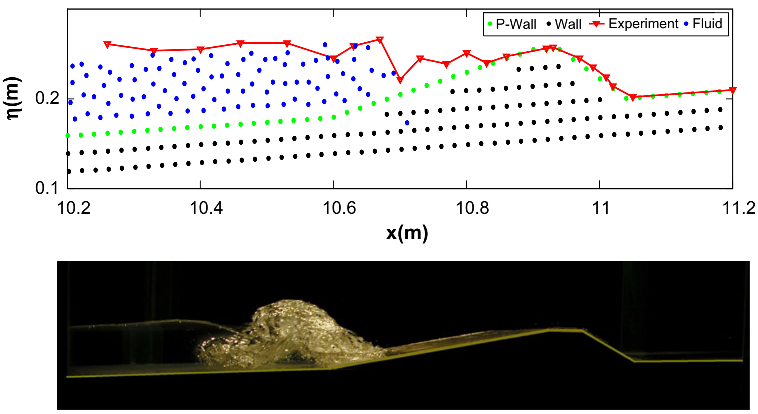
The wave caused by the overtopping tsunami causes severe damage at the landward of the seawall, which represents the most typical damage mechanism for seawalls (Suppasri et al. 2012). It is very important to evaluate the overtopping discharge, to get better understanding of the TSI. There is a lack of experimental data for the overtopping volume, so we compared the numerical results with the result of an empirical formula,

$$Q = 0.55\sqrt{gh_c^3}, \tag{19}$$

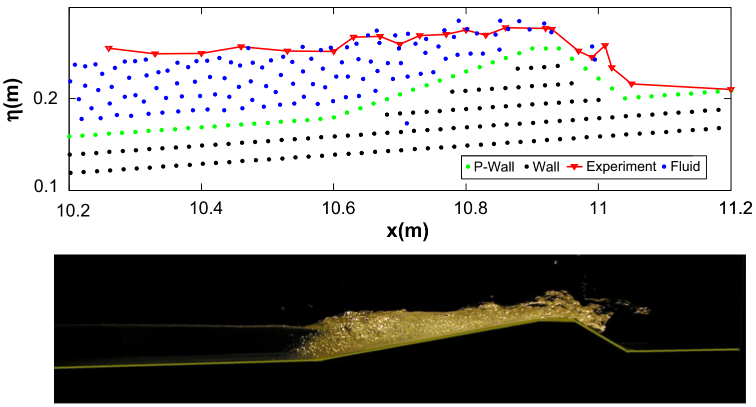
where Q is the overtopping volume, h_c is the overtopping height, and 0.55 is an empirical value (Yamamoto et al. 2006). The overtopping height h_c was estimated by averaging the



(a) t=2.89

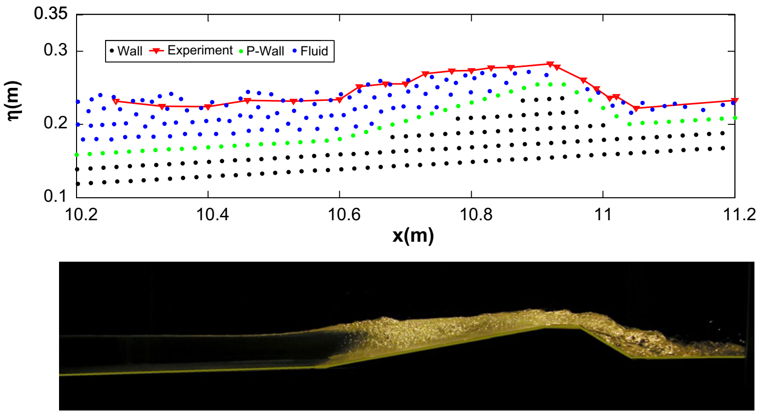


(b) t=3.01

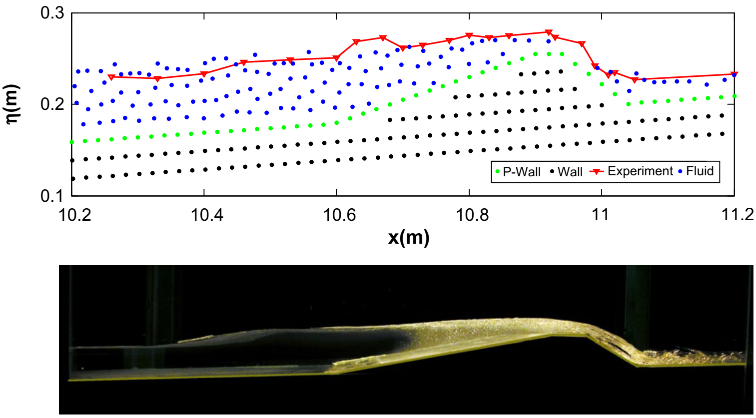


(c) t=3.19

Fig. 4 Comparison of the free-surface elevation from the simulated results and the experimental data (the experimental data and photographs were provided by Hsiao and Lin 2010)



(d) t=3.35



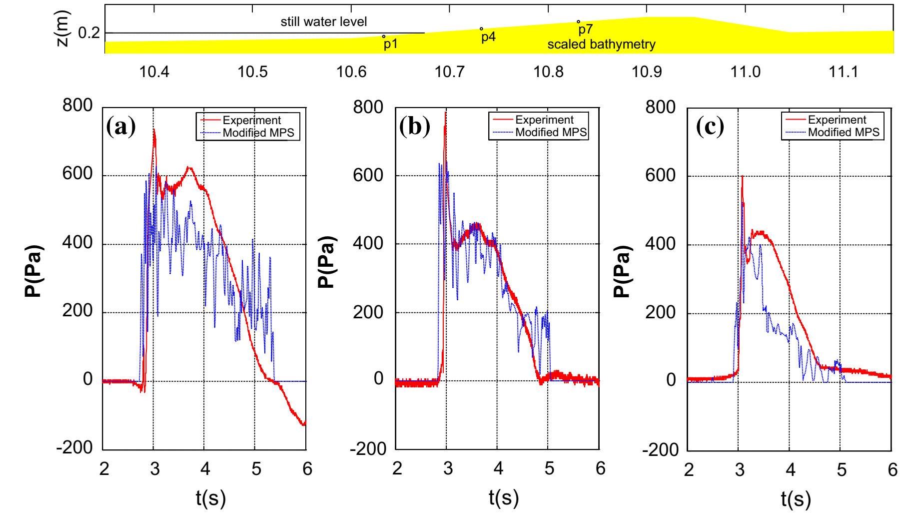
(e) t=3.71

Fig. 4 continued

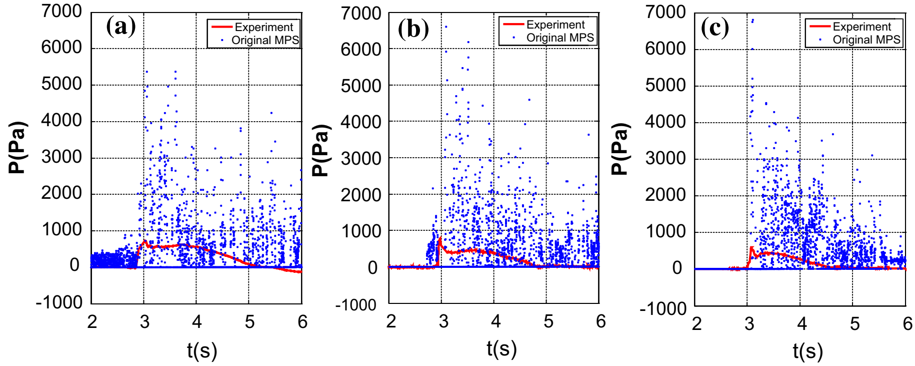
elevations of a tsunami wave, measured by two wave gauges placed on the crown of the seawall. We found that h_c was 0.02216 m and then Q was 5.6798×10^{-3} m^3/m/s. A counter was inserted into the code to record the number of particles passing over the seawall and estimate the overtopping discharge in the numerical simulation. We found that the overtopping volume was 4.2449×10^{-3} m^3/m/s. The numerical result was consistent with the empirical formula. This result is expected, because of the accuracy of the surface elevation. Any deviations in the overtopping volume may be because of the limited information provided for the calculation Q using the empirical formulae. As previously mentioned, the overtopping volume estimate is acceptable considering the limited information.

6.5 Analysis of the simulation results

Our comparison of the calculated results and the experiment data considered the wave height, free-surface elevation, dynamic wave pressure, and overtopping volume,



(1) The comparison of experimental pressure and calculated pressure by modified MPS



(2) The comparison of experimental pressure and calculated pressure by original MPS

Fig. 5 Time series of the pressure at **a** P1; **b** P4; and **c** P7 (the experimental data were provided by Hsiao and Lin 2010)

all of which are essential for the design of a coastal structure. We found that modified MPS results were very similar to the physical model tests. The comparisons indicated that the modified MPS method can simulate the TSI and qualitatively and quantitatively reproduce the whole process of the tsunami approaching and overtopping the seawall.

Further calibration and validation of the numerical calculations of the TSI are needed. It should be pointed out that, in reality, the coastal structure will be excessively deformed or destroyed by the tsunami impact. However, the modified MPS method can sufficiently model the TSI and capture the essential dynamic characteristics. As a result, the simulation aids our understanding of the TSI features and provides preliminary suggestions for the design of coastal structures.

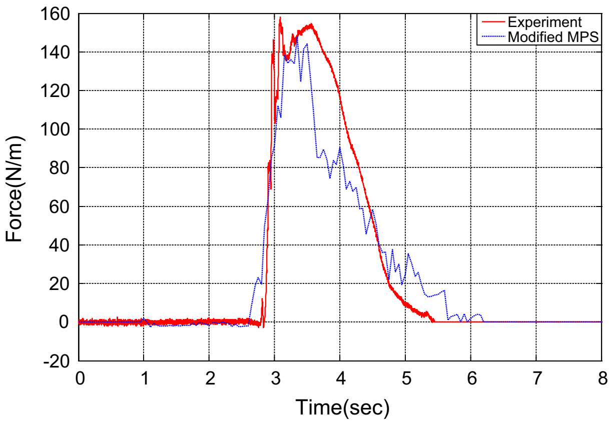


Fig. 6 Time series of the wave force on the seaward slope (the experimental data were provided by Hsiao and Lin 2010)

7 Conclusions

The MPS method is applied to study the dynamic features of the TSI considering the difficulties of the traditional mesh-based method when describing large deformations. The time series of the pressure is fundamental to the design of coastal structures. We calculated this time series by improving three aspects of the original MPS code including the kernel function, the source term of the Poisson equation, and the search for the free-surface particles. These modifications reduced the pressure fluctuations. We verified the improvement of the modified MPS method by applying it to two benchmark cases: the dam break and static pressure problems. The simulation results were compared with the experiment data or the theoretical solution and showed good agreement.

We then applied the modified MPS method to simulate the TSI, using a solitary wave to model the tsunami wave. We considered some essential dynamic characteristics, including the wave height, free-surface elevation, dynamic wave pressure, and overtopping volume. We found that the numerical results agreed well with the experimental data, which indicates that the modified MPS method can capture the essential dynamic features and reproduce the entire TSI process.

Considering the complexity of the TSI, further calibration and validation are needed to improve the accuracy and robustness of the simulation. However, we firmly believe that the current computation can help a designer to understand the fundamental dynamic characteristic of the TSI and can provide preliminary suggestions on the design of coastal structures. This could reduce or even prevent similar disasters.

Acknowledgments The authors would like to express their gratitude to Prof. Shih-Chun Hsiao and Dr. Ting-Chieh Lin (National Cheng Kung University) for providing the original experimental data.

References

Arikawa T, Ishikawa N, Beppu M, Tatesawa H (2012) Collapse mechanisms of seawall due to the March 2011 Japan tsunami using the MPS method. Int J Prot Struct 3(4):457–476

Boussinesq JV (1872) Théorie des ondes et des remous qui se propagent le long d'un canal rectangulaire horizontal, en communiquant au liquide contenu dans ce canal des vitesses sensiblement pareilles de la surface au fond. J Math Pures Appl 17(2):55–108

Bullock GN, Crawford AR, Hewson PJ, Walkden MJA, Bird PAD (2001) The influence of air and scale on wave impact pressures. Coast Eng 42(4):291–312
Cleary PW, Prakash M (2004) Discrete-element modelling and smoothed particle hydrodynamics: potential in the environmental sciences. Philos Trans R Soc Lond Ser A 362(1822):2003–2030
Farreras SF (2000) Post-tsunami field survey procedures: an outline. Nat Hazards 21(2–3):207–214
Fujima K (2006) Effect of a submerged bay-mouth breakwater on tsunami behavior analyzed by 2D/3D hybrid model simulation. Nat Hazards 39(2):179–193
Garcia N, Lara JL, Losada IJ (2004) 2-D numerical analysis of near-field flow at low-crested permeable breakwaters. Coast Eng 51(10):991–1020
Goring DG (1979) Tsunamis: the propagation of long waves onto A Shelf, Ph.D. Thesis, California Institute of Technology
Gotoh H, Shao S, Memita T (2004) SPH-LES model for numerical investigation of wave interaction with partially immersed breakwater. Coast Eng J 46(1):39–63
Gotoh H, Ikari H, Memita T, Sakai T (2005) Lagrangian particle method for simulation of wave overtopping on a vertical seawall. Coast Eng J 47(2&3):157–181
Hanzawa M, Matsumoto A, Tanaka H (2012) Applicability of CADMAS-SURF to evaluate detached breakwater effects on solitary tsunami wave reduction. Earth Planets Space 64(10):955–964
Hayashi M, Gotoh H, Memita T, Sakai T (2001) Gridless numerical analysis of wave breaking and overtopping at upright seawall. In: 27th international conference of coastal engineering, Sydney, pp 2100–2113
Heo S, Koshizuka S, Oka Y (2002) Numerical analysis of boiling on high heat-flux and high subcooling condition using MPS-MAFL. Int J Heat Mass Transf 45(13):2633–2642
Hsiao SC, Lin TC (2010) Tsunami-like solitary waves impinging and overtopping an impermeable seawall: experiment and RANS modeling. Coast Eng 57(1):1–18
Huang Y, Zhu C (2014) Simulation of flow slides in municipal solid waste dumps using a modified MPS method. Nat Hazards 74(2):491–508
Jung SJ, Park JC, Lee BH, Ryu MC, Kim YS (2008) Numerical simulation of two-dimensional floating body motion in waves using particle method. Korean Soc Ocean Eng 22(2):20–27 (in Korean)
Khayyer A, Gotoh H (2009) Modified moving particle semi-implicit methods for the prediction of 2D wave impact pressure. Coast Eng 56(4):419–440
Kondo M, Koshizuka S (2011) Improvement of stability in moving particle semi-implicit method. Int J Numer Methods Fluids 65(6):638–654
Koshizuka S, Oka Y (1996) Moving-particle semi-implicit method for fragmentation of incompressible fluid. Nucl Sci Eng 123(3):421–434
Koshizuka S, Go S, Oka Y, and Tanizawa K (2004) Numerical analysis of two-dimensional experiment of shipping water on deck using a particle method. In: Proceedings of the 2004 ASME heat transfer/fluid engineering summer conference, Charlotte, HT-FED04-56293
Lau TL, Ohmachi T, Inoue S, Lukkunaprasit P (2011) Experimental and numerical modeling of tsunami force on bridge decks. In: Mokhtari M (ed) Tsunami—a growing disaster. InTech, 6:105–130
Lee BH, Park JC, Kim MH, Hwang SC (2011) Step-by-step improvement of MPS method in simulating violent free-surface motions and impact-loads. Computer Methods Appl Mech Eng 200(9):1113–1125
Lin P (2004) A numerical study of solitary wave interaction with rectangular obstacles. Coast Eng 51(1):35–51
Lin JZ, Liang GZ, Zhao Y, Xie MF (1980) Focal mechanism and tectonic stress field of coastal Southeast China. Acta Seismol Sin 2(3):245–257 (in Chinese)
Losada IJ, Lara JL, Guanche R, Gonzalez-Ondina JM (2008) Numerical analysis of wave overtopping of rubble mound breakwaters. Coast Eng 55(1):47–62
National Natural Science Foundation of China (2011) Water conservancy and ocean engineering science development strategy report. Science Press, Beijing (in Chinese)
Ohyama T, Nadaoka K (1992) Modeling the transformation of nonlinear waves passing over a submerged dike. Coast Eng Proc 1(23):526–539
Rogers BD, Dalrymple RA, Stansby PK (2010) Simulation of caisson breakwater movement using 2-D SPH. J Hydraul Res 48(S1):135–141
Shao S, Lo EYM (2003) Incompressible SPH method for simulating newtonian and non-newtonian flows with a free surface. Adv Water Resour 26(7):787–800
State Council Information Office of China (1998) The development of China's marin programs. State Council Information Office of China, Beijing
Suppasri A, Koshimura S, Imai K, Mas E, Gokon H, Muhari A, Imamura F (2012) Damage characteristic and field survey of the 2011 great East Japan tsunami in miyagi prefecture. Coast Eng J 54(1):1250005 (Special Issue of 2011 Tohoku tsunami)

Suzuki Y, Koishikawa M, Koshizuka S, Okamoto T, Kaneko N, Takamatsu A, Fujii T (2006) Numerical simulation of adhesion of cells in micro channels using the MPS method. Jpn Soc Mech Eng Part B 18(9):2109–2116 (in Japanese)

Synolakis CE, Bernard EN (2006) Tsunami science before and beyond boxing day 2004. Philos Trans R Soc A: Math Phys Eng Sci 364(1845):2231–2265

Tanaka M, Masunaga T (2010) Stabilization and smoothing of pressure in MPS method by quasi-compressibility. J Comput Phys 229(11):4279–4290

Titov VV, Gonzalez FI, Bernard EN, Eble MC, Mofjeld HO, Newman JC, Venturato AJ (2005) Real-time tsunami forecasting: challenges and solutions. Nat Hazard 35:41–58

Tsubota KI, Wada S, Kamada H, Kitagawa Y, Lima R, Yamaguchi T (2006) A particle method for blood flow simulation: application to flowing red blood cells and platelets. J Earth Simul 5:2–7

Ward SN (1989) Tsunami. In: James DE (ed) The encyclopedia of solid earth geophysics. Van Nostrand Reinhold, Hoboken, pp 1279–1292

Yamamoto Y, Takanashi H, Tomita T (2006) Mechanism of coastal structure damage due to tsunamis, and case studies of coastal damage from Indian Ocean tsunami. Proceedings of 30th international conference on coastal engineering. San Diego, pp 5126–5138

Zhou QH, Adams WM (1988) Tsunami risk analysis for China. Nat Hazards 1:181–195